QPCode

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP BCA (HONOURS) EXAMINATION OCTOBER 2026

FIFTH SEMESTER
DSE- MG5DSEBCA305- NATURAL LANGUAGE PROCESSING
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. What is meant by Natural Language Processing (NLP)? [K] / [CO1]
2. What is articulatory phonetics? [U] / [CO1]
3. What is a Finite State Automaton? [K] / [CO2]
4. Define a lexeme and a phoneme. [K] / [CO3]
5. Give the significance of text coherence? [U] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. Explain the different levels of language analysis in NLP. [K] / [CO1]
7. What is WordNet? Mention its main purpose in NLP. [U] / [CO1]
8. Describe stochastic Part-of-Speech tagging using Hidden Markov Models. [U] / [CO2]
9. Explain Context-Free Grammar (CFG) and construct a simple CFG for generating basic English sentences. [A] / [CO3]
10. Explain the structure and applications of TreeBank in natural language processing. [U] / [CO3]
11. Given the sentence "The student went to the bank to deposit money," explain how context helps in identifying the correct sense of the word *bank*. [A] / [CO4]
12. Analyse how pronoun resolution, cohesion, and coherence help an NLP system understand the structure and meaning of a text. [An]/[CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13. Explain Finite State Transducers (FSTs) in detail. Describe their role in morphological analysis and generation with a suitable example. [An] / [CO2]

14. Describe the major parsing techniques used in NLP. Discuss top-down parsing, bottom-up parsing, chart parsing, unification-based parsing, and probabilistic parsing using PCFGs. [An] / [CO3]

15. Explain the major approaches to Word Sense Disambiguation. Discuss dictionary-based and machine learning approaches and the evaluation of WSD systems. [U] / [CO4]

[15x2 = 30]